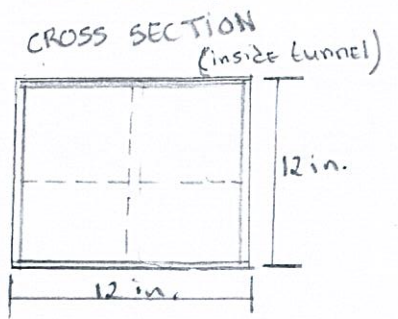
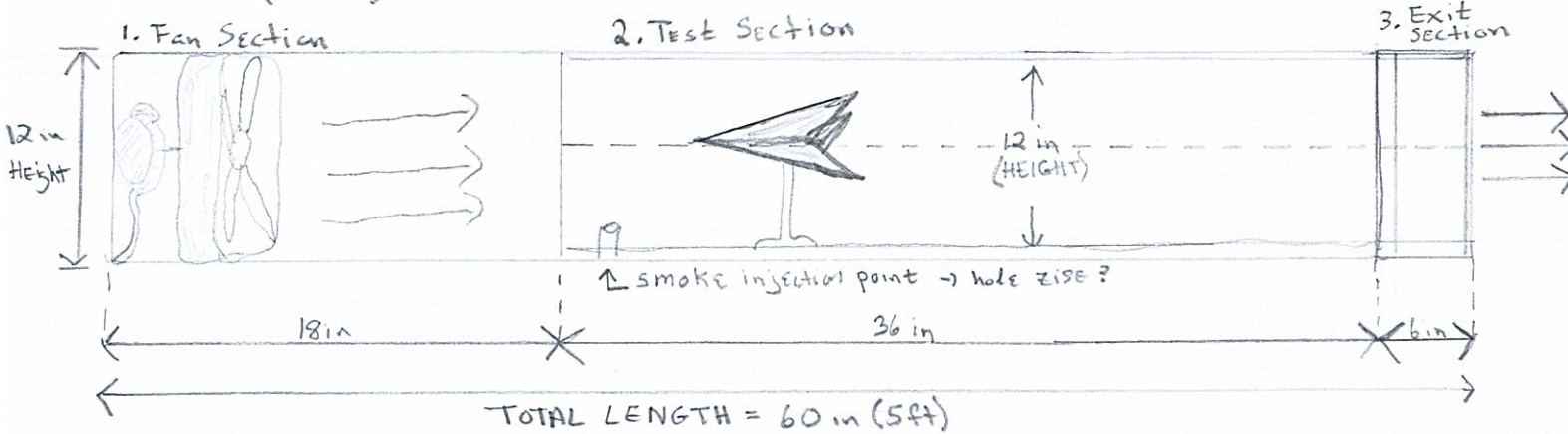


SIDE VIEW (Looking from the side)



Fan Section: 18 in.

Test Section: 36 in.

Exit Section: 6 in.

Total Length: 60 in.

Gross Section: 12 in x 12 in

Calculate Tunnel Volume:

Total Length: 60 in

Convert:  $\frac{60}{12} = 5 \text{ ft}$

Volume:  $V = L \times W \times H$

$V = (5)(1)(1)$

$V = 5 \text{ ft}^3$

Total Volume = 5 cubic feet

Air Velocity Equation:

$$\text{Velocity} = \frac{\text{CFM}}{\text{Area}} \rightarrow V = \frac{\text{CFM}}{1 \text{ ft}^2}$$

\* Velocity = CFM \*

Convert ft/min → mph

1 mile = 5280 ft

1 hour = 60 min

1 mph =  $\frac{5280}{60}$

1 mph = 88 ft/min

mph =  $\frac{\text{ft/min}}{88}$

For this Tunnel: ft/min = CFM

mph =  $\frac{\text{CFM}}{88}$

Calculate Cross-Sectional Area: inches → feet

$12 \text{ in} = \frac{12}{12} = 1 \text{ ft}$

Area = (1 ft)(1 ft)

$A = 1 \text{ ft}^2$

Gross Sectional Area = 1 ft<sup>2</sup>

# Different Types of Fans Air Velocity Calculations

Fan 1 - # 7663N17

$$\text{CFM} = 590$$

$$\text{Velocity} = \frac{590}{1}$$

$$\text{Velocity} = 590 \text{ ft/min}$$

Convert to mph:

$$\text{mph} = \frac{590}{88}$$

$$\text{mph} = 6.70$$

Air Changes:

$$\text{Tunnel Volume} = 5 \text{ ft}^3$$

$$\frac{590}{5} = 118$$

118 air changes/minute

Fan 2 - # 9191K37

$$\text{CFM} = 600$$

$$\text{Velocity} = \frac{600}{1}$$

$$\text{Velocity} = 600 \text{ ft/min}$$

Convert to mph:

$$\text{mph} = \frac{600}{88}$$

$$\text{mph} = 6.82$$

Air Changes:

$$\text{Tunnel Volume} = 5 \text{ ft}^3$$

$$\frac{600}{5} = 120$$

120 air changes/minute

Fan 3 - 9191K82

$$\text{CFM} = 600$$

$$\text{Velocity} = \frac{600}{1}$$

$$\text{Velocity} = 600 \text{ ft/min}$$

Convert to mph:

$$\text{mph} = \frac{600}{88}$$

$$\text{mph} = 6.82$$

Air Changes:

$$\text{Tunnel Volume} = 5 \text{ ft}^3$$

$$\frac{600}{5} = 120$$

120 air changes/minute

Fan 4 - 1932K32

$$\text{CFM} = 800$$

$$\text{Velocity} = \frac{800}{1}$$

$$\text{Velocity} = 800 \text{ ft/min}$$

Convert to mph:

$$\text{mph} = \frac{800}{88}$$

$$\text{mph} = 9.09 \text{ mph}$$

Air Changes:

$$\text{Tunnel Volume} = 5 \text{ ft}^3$$

$$\frac{800}{5} = 160$$

160 air changes/minute



## Different Types of Fans Air Velocity Calculations Cont.

Fan 5 - 7663N22

$$\text{CFM} = 1000$$

$$\text{Velocity} = \frac{1000}{1}$$

$$\text{Velocity} = 1000 \text{ ft/min}$$

Convert to mph:

$$\text{mph} = \frac{1000}{88}$$

$$\text{mph} = 11.36 \text{ mph}$$

Air Changes:

$$\text{Tunnel Volume} = 5 \text{ ft}^3$$

$$\frac{1000}{5} = 200$$

200 air changes/minute

Fan 6 - 7663N24

$$\text{CFM} = 1300$$

$$\text{Velocity} = \frac{1300}{1}$$

$$\text{Velocity} = 1300 \text{ ft/min}$$

Convert to mph:

$$\text{mph} = \frac{1300}{88}$$

$$\text{mph} = 11.36 \text{ mph}$$

Air Changes:

$$\text{Tunnel Volume} = 5 \text{ ft}^3$$

$$\frac{1300}{5} = 260$$

260 air changes/minute

Fan 7 - 2138K12

$$\text{CFM} = 1500$$

$$\text{Velocity} = \frac{1500}{1}$$

$$\text{Velocity} = 1500 \text{ ft/min}$$

Convert to mph:

$$\text{mph} = \frac{1500}{88}$$

$$\text{mph} = 17.05$$

Air Changes:

$$\text{Tunnel Volume} = 5 \text{ ft}^3$$

$$\frac{1500}{5} = 300$$

300 air changes/minute

# Fans Comparison Table

| Fan     | CFM  | Velocity ft/min | Velocity (mph) | Air Changes/min |
|---------|------|-----------------|----------------|-----------------|
| 7663N17 | 590  | 590             | 6.70           | 118             |
| 9191K37 | 600  | 600             | 6.82           | 120             |
| 9191K82 | 600  | 600             | 6.82           | 120             |
| 1932K32 | 800  | 800             | 9.09           | 160             |
| 7663N22 | 1000 | 1000            | 11.36          | 200             |
| 7663N24 | 1300 | 1300            | 14.77          | 260             |
| 2138K12 | 1500 | 1500            | 17.05          | 300             |

## Fan Selection Analysis & Recommendation

After Revising the wind tunnel design to a simplified configuration consisting of a fan section, test section, & exit section, I performed airflow calculations for SEVEN potential fan options from McMaster-Carr.

- Wind Tunnel has a constant cross-sectional area of  $12\text{in} \times 12\text{in} = 1\text{ft}^2$   
 & a total volume of  $5\text{ft}^3$

Using the continuity equation:  $\text{Velocity} = \frac{\text{CFM}}{\text{Area}}$

→ the airflow velocity inside the test section is numerically (=) to the fans airflow rating in feet per minute because the test section is exactly  $1\text{ft}^2$ .

Airflow Fans of (590-600 CFM)

- produced air speed of approximately 6.7-6.8 mph
- would likely provide smooth airflow & good smoke visualization.
- May not generate enough aerodynamics effects to clearly distinguish differences between paper airplane designs.

Continue...



# Fan Selection Analysis & Recommendation - Cont.

## Air flow fans (800-1000 CFM)

- appears to provide the best balance between airflow strength, flow visualization, & experimental control.
- produced airspeeds between approximately 9.09 - 11.36 mph.

## Air flow fans (1300-1500 CFM)

- produced airspeeds between approximately 14.8 - 17.1 mph.
- would likely create stronger wake patterns & more visible airflow effects
- may also introduce additional turbulence & make smoke visualization more difficult without flow-conditioning devices.

## Conclusion:

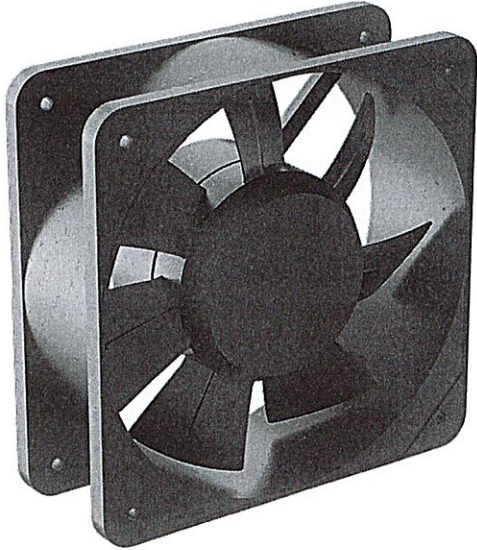
Based on the preliminary airflow calculations, the 1000 CFM McMaster 7663N22 fan appears to be the best overall choice for the current scope of the project, providing an estimated test-section airspeed of approximately 11.4 mph, which should be sufficient for observing airflow attachment, wake formation, and aerodynamic differences between paper airplane designs.

For the goals of this project, paper airplane airflow visualization, smoke testing, student research, and the planned September paper airplane competition. The 1000 CFM fan currently appears to offer the best balance of airflow quality, visualization capability, cost, and experimental control.



## High-Pressure Washdown Equipment-Cooling Fan

115V AC, 205 mm High x 205 mm Wide x 72 mm Deep Overall

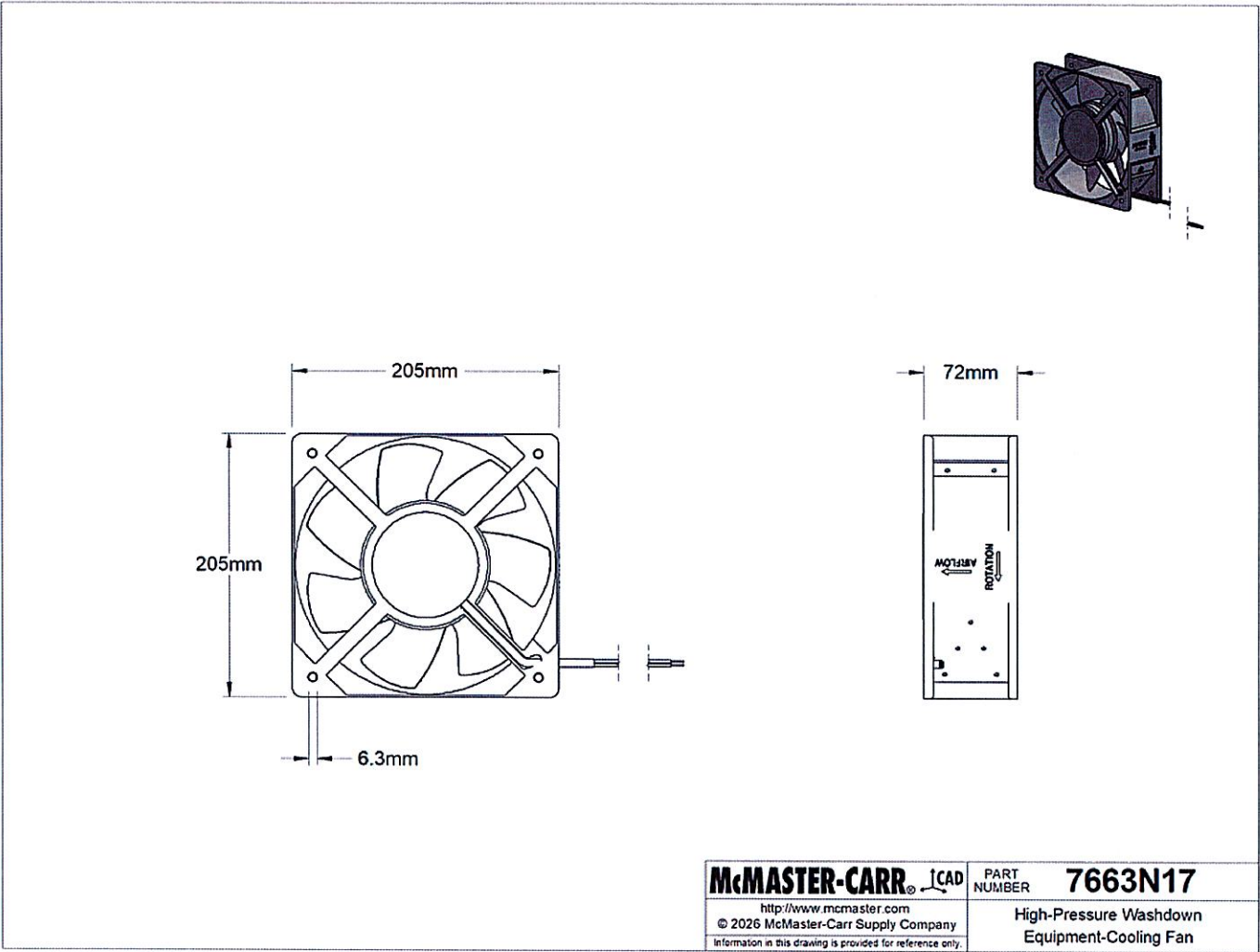


Delivers tomorrow

\$192.49 each  
7663N17

|                                                 |                                                       |
|-------------------------------------------------|-------------------------------------------------------|
| Shape                                           | Square                                                |
| Frame Material                                  | Aluminum                                              |
| Blade Material                                  | Steel                                                 |
| Voltage                                         | 115V AC                                               |
| Electrical Phase                                | Single                                                |
| Frequency                                       | 60 Hz                                                 |
| Electrical Connection                           | Wire Leads                                            |
| Overall                                         |                                                       |
| Height                                          | 205 mm                                                |
| Width                                           | 205 mm                                                |
| Depth                                           | 72 mm                                                 |
| Airflow                                         | 590 ft <sup>3</sup> /min                              |
| Volume (Sound)                                  | 67 dBA                                                |
| Current                                         | 0.67 amp                                              |
| Mounting                                        |                                                       |
| Fasteners Included                              | No                                                    |
| Hole Diameter                                   | 6.3 mm                                                |
| Enclosure Rating                                | IP69K                                                 |
| Certification                                   | CE Marked, UL Recognized Component                    |
| Application                                     | Cooling                                               |
| Environment                                     | Dusty, High-Pressure Washdown, Washdown, Wet          |
| Power Source                                    | Electric                                              |
| Thermostat Included                             | No                                                    |
| Country of Origin                               | Taiwan                                                |
| DFARS Compliance                                | Specialty Metals COTS-Exempt                          |
| Export Control Classification Number (ECCN)     | EAR99                                                 |
| REACH Compliance                                | REACH (EC 1907/2006) (06/27/2024, 241 SVHC) Compliant |
| RoHS Compliance                                 | RoHS 3 (2015/863/EU) Compliant                        |
| Schedule B Number                               | 841459.9080                                           |
| U.S.-Mexico-Canada Agreement (USMCA) Qualifying | No                                                    |

Cool down electronics in areas that require frequent and intense sanitizing. Rated IP69K, these AC-powered fans withstand harsh jets of water and block dust from reaching their motor. All have ball bearings that reduce friction and heat from operation, extending their life. Also known as muffin fans.



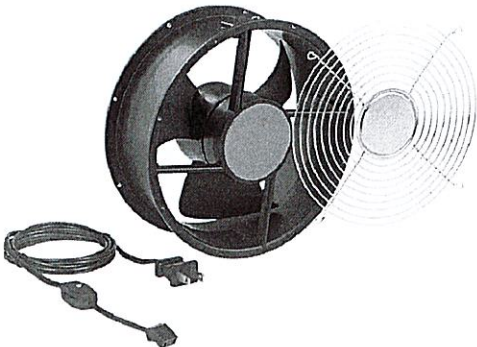
The information in this 3-D model is provided for reference only.





Equipment-Cooling Fan Kit

Steel Guard and Switch, 120V AC, 10" Diameter



Delivers tomorrow

\$121.67 each  
9191K37

|                                                 |                                                       |
|-------------------------------------------------|-------------------------------------------------------|
| Shape                                           | Round                                                 |
| Guard Material                                  | Steel                                                 |
| Includes                                        | Fan, Guard, On/Off Switch, Power Cord                 |
| Voltage                                         | 120V AC                                               |
| Electrical Phase                                | Single                                                |
| Frequency                                       | 50 Hz/60 Hz                                           |
| Electrical                                      |                                                       |
| Connector Component                             | Plug                                                  |
| Number of Blades                                | 2                                                     |
| Overall                                         |                                                       |
| Diameter                                        | 10.000" (254 mm)                                      |
| Depth                                           | 3.504" (89 mm)                                        |
| Airflow                                         | 600 ft³/min                                           |
| Temperature Range                               | 0° F to 155° F                                        |
| Volume (Sound)                                  | 49 dBA                                                |
| Current                                         | 0.24 amp                                              |
| Frame Material                                  | Aluminum                                              |
| Mounting                                        |                                                       |
| Fasteners Included                              | No                                                    |
| Hole Diameter                                   | 0.157" (4 mm)                                         |
| Application                                     | Cooling                                               |
| Blade Material                                  | Plastic                                               |
| Certification                                   | UL Listed                                             |
| Cord Length                                     | 6 ft.                                                 |
| Electrical Connection                           | Straight Blade                                        |
| Power Source                                    | Electric                                              |
| Thermostat Included                             | No                                                    |
| Country of Origin                               | United States                                         |
| DFARS Compliance                                | Specialty Metals COTS-Exempt                          |
| Export Control Classification Number (ECCN)     | EAR99                                                 |
| REACH Compliance                                | REACH (EC 1907/2006) (02/04/2026, 253 SVHC) Compliant |
| RoHS Compliance                                 | RoHS 3 (2015/863/EU) Compliant                        |
| Schedule B Number                               | 841459.9080                                           |
| U.S.–Mexico–Canada Agreement (USMCA) Qualifying | Yes                                                   |

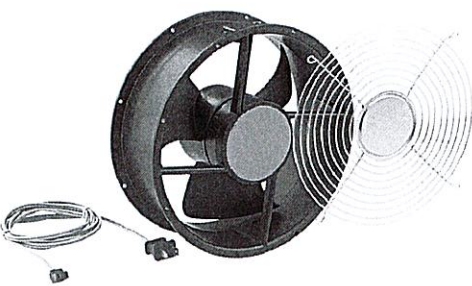
Kits include a fan, guard, and power cord. They keep heat-sensitive equipment such as electronic components cool.





Equipment-Cooling Fan Kit

with Thermostat, Round, 600 CFM Airflow, Steel Guard



Delivers tomorrow

\$114.49 each

9191K82

|                                                 |                                                       |
|-------------------------------------------------|-------------------------------------------------------|
| Shape                                           | Round                                                 |
| Guard Material                                  | Steel                                                 |
| Includes                                        | Fan, Guard, Power Cord                                |
| Voltage                                         | 120V AC                                               |
| Electrical Phase                                | Single                                                |
| Frequency                                       | 50 Hz/60 Hz                                           |
| Electrical                                      |                                                       |
| Connector Component                             | Plug                                                  |
| Number of Blades                                | 2                                                     |
| Overall                                         |                                                       |
| Diameter                                        | 10.000" (254 mm)                                      |
| Depth                                           | 3.504" (89 mm)                                        |
| Airflow                                         | 600 ft <sup>3</sup> /min                              |
| Temperature Range                               | 0° F to 155° F                                        |
| Volume (Sound)                                  | 49 dBA                                                |
| Current                                         | 0.24 amp                                              |
| Frame Material                                  | Aluminum                                              |
| Thermostat Temperature Measurement Range        | 75° F to 86° F                                        |
| Mounting                                        |                                                       |
| Fasteners Included                              | No                                                    |
| Hole Diameter                                   | 0.157" (4 mm)                                         |
| Application                                     | Cooling                                               |
| Blade Material                                  | Plastic                                               |
| Certification                                   | UL Recognized Component                               |
| Cord Length                                     | 6 ft.                                                 |
| Electrical Connection                           | Straight Blade                                        |
| Power Source                                    | Electric                                              |
| Thermostat Control                              | Fixed                                                 |
| Thermostat Included                             | Yes                                                   |
| Country of Origin                               | United States                                         |
| DFARS Compliance                                | Specialty Metals COTS-Exempt                          |
| Export Control Classification Number (ECCN)     | EAR99                                                 |
| REACH Compliance                                | REACH (EC 1907/2006) (02/04/2026, 253 SVHC) Compliant |
| RoHS Compliance                                 | RoHS 3 (2015/863/EU) Compliant                        |
| Schedule B Number                               | 841459.9080                                           |
| U.S.–Mexico–Canada Agreement (USMCA) Qualifying | Yes                                                   |

Thermostat allows the fans in these kits to switch on only when needed. All kits include a fan, guard, and power cord. They keep heat-sensitive

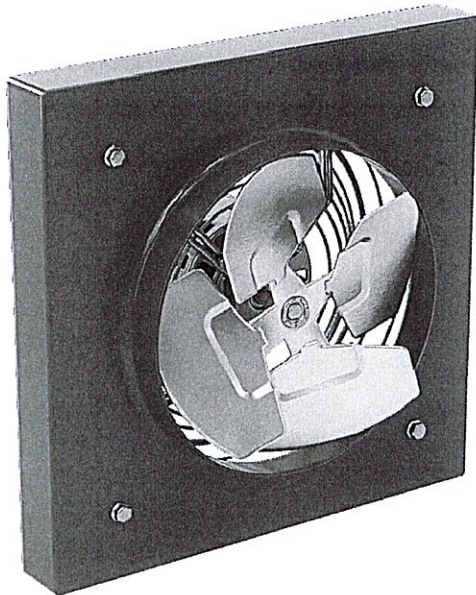






## Direct-Drive Wall-Mount Exhaust Fan

120V AC, Open Dripproof, 10" Blade Diameter



Delivers in 2 weeks

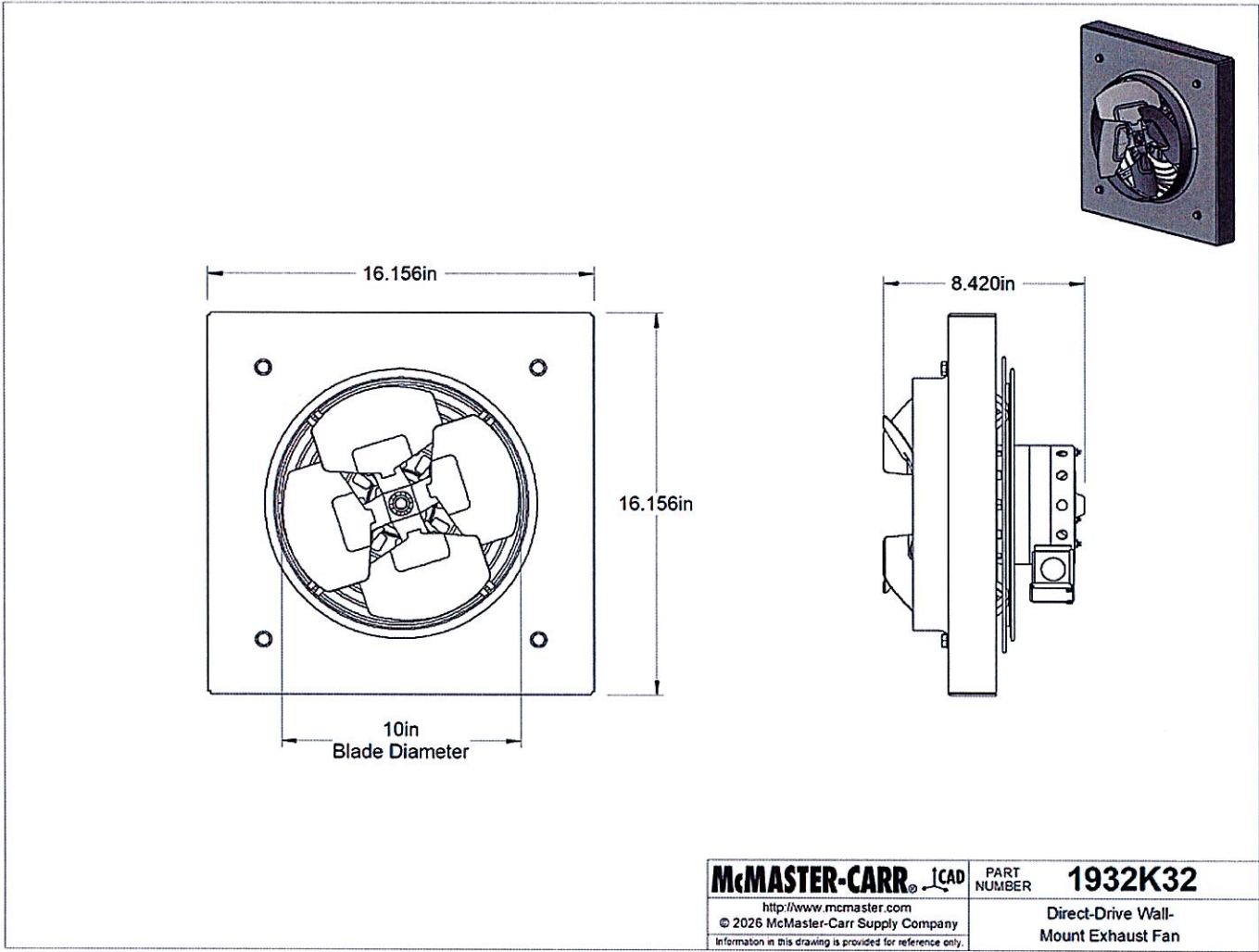
\$553.58 each  
1932K32

|                                                 |                                     |
|-------------------------------------------------|-------------------------------------|
| Voltage                                         | 120V AC                             |
| Electrical Phase                                | Single                              |
| Electrical Connection Type                      | Hardwire                            |
| Motor Enclosure Type                            | Open Dripproof (ODP)                |
| Blade Diameter                                  | 10"                                 |
| Airflow                                         | 800 ft <sup>3</sup> /min            |
| Volume (Sound) @ Distance                       | 57 dBA @ 5 ft.                      |
| Drive Type                                      | Direct                              |
| Motor Speed                                     | 1,550 rpm                           |
| Number of Speeds                                | 1                                   |
| Power                                           | 1/20 hp                             |
| Overall                                         |                                     |
| Height                                          | 16 1/8"                             |
| Width                                           | 16 1/8"                             |
| Depth                                           | 8 3/8"                              |
| Ambient Temperature                             | 100° F                              |
| Frame Material                                  | Powder-Coated Steel                 |
| Blade Material                                  | Aluminum                            |
| Certification                                   | C-UL Listed, UL Listed              |
| Current                                         | 1.7 amp                             |
| Mounting Location                               | Wall                                |
| Power Source                                    | Electric                            |
| UL File Number                                  | E35337                              |
| Country of Origin                               | United States                       |
| DFARS Compliance                                | Specialty Metals COTS-Exempt        |
| Export Control Classification Number (ECCN)     | EAR99                               |
| REACH Compliance                                | Not Compliant                       |
| RoHS Compliance                                 | Not Compliant                       |
| Schedule B Number                               | 841459.9080                         |
| U.S.-Mexico-Canada Agreement (USMCA) Qualifying | Yes                                 |
| Related Product                                 | <a href="#">Exhaust Fan Collars</a> |

Direct-drive fans have fewer moving parts than belt-drive fans for less maintenance. Use these fans to draw stagnant and humid air out of a room or area.

**Open Dripproof (ODP) Motor Enclosure**—Fans with an open dripproof (ODP) motor enclosure are for relatively clean, dry, indoor locations.

**Exhaust Fan Collars**—Use a collar (sold separately) to connect these fans to a [louver](#).

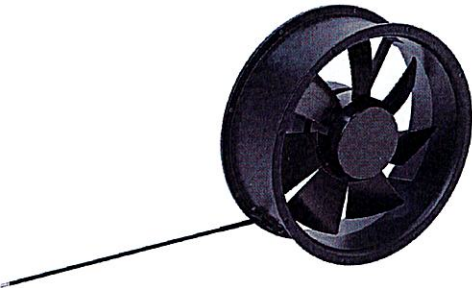


The information in this 3-D model is provided for reference only.



High-Pressure Washdown Equipment-Cooling Fan

115V AC, 254 mm Diameter x 89 mm Deep Overall



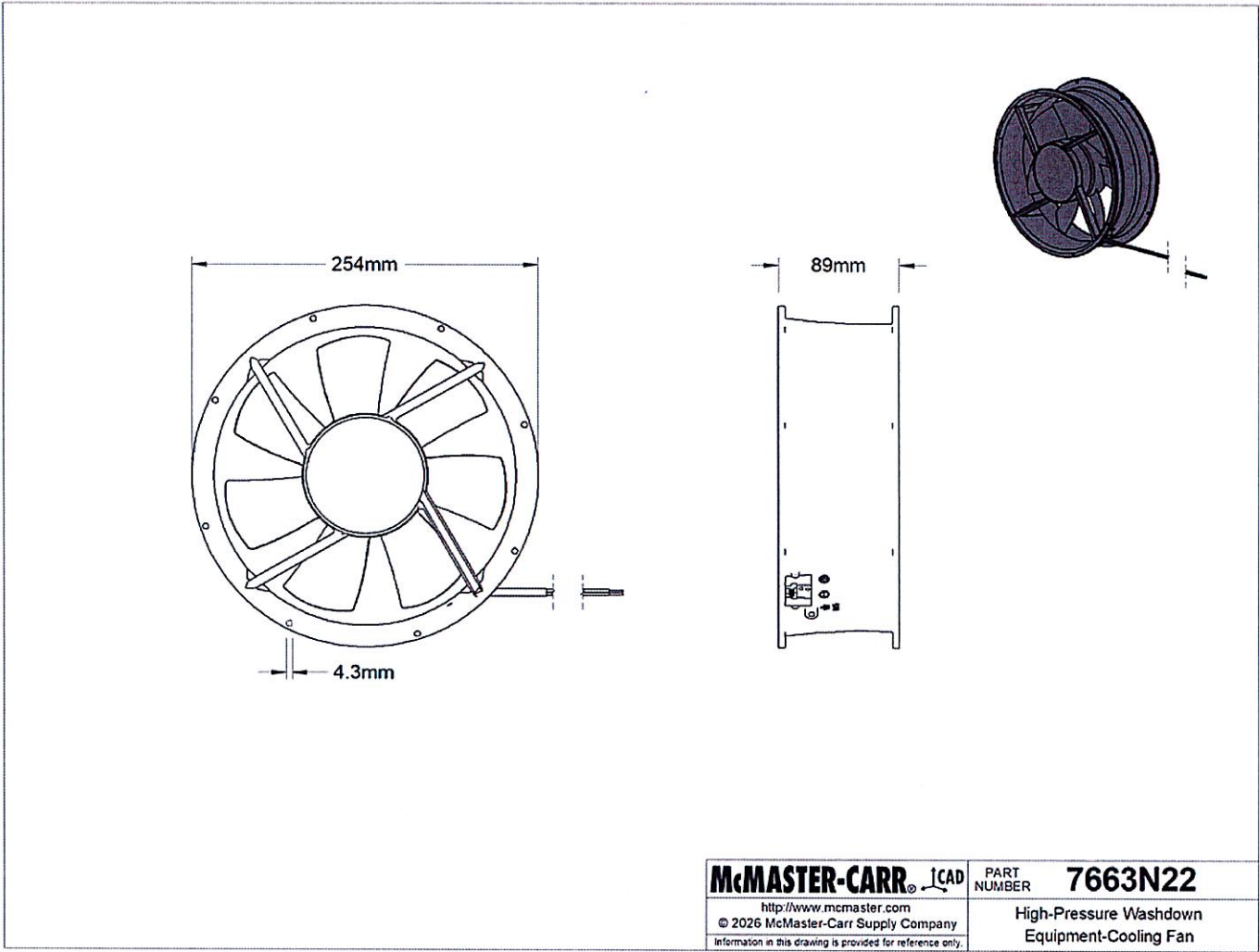
Delivers tomorrow

\$226.68 each  
7663N22

|                                                 |                                                       |
|-------------------------------------------------|-------------------------------------------------------|
| Shape                                           | Round                                                 |
| Frame Material                                  | Aluminum                                              |
| Blade Material                                  | Steel                                                 |
| Voltage                                         | 115V AC                                               |
| Electrical Phase                                | Single                                                |
| Frequency                                       | 60 Hz                                                 |
| Electrical Connection                           | Wire Leads                                            |
| Overall                                         |                                                       |
| Diameter                                        | 254 mm                                                |
| Depth                                           | 89 mm                                                 |
| Airflow                                         | 1,000 ft <sup>3</sup> /min                            |
| Volume (Sound)                                  | 70 dBA                                                |
| Current                                         | 0.87 amp                                              |
| Mounting                                        |                                                       |
| Fasteners Included                              | No                                                    |
| Hole Diameter                                   | 4.3 mm                                                |
| Enclosure Rating                                | IP69K                                                 |
| Certification                                   | CE Marked, UL Recognized Component                    |
| Application                                     | Cooling                                               |
| Environment                                     | Dusty, High-Pressure Washdown, Washdown, Wet          |
| Power Source                                    | Electric                                              |
| Thermostat Included                             | No                                                    |
| Country of Origin                               | Taiwan                                                |
| DFARS Compliance                                | Specialty Metals COTS-Exempt                          |
| Export Control Classification Number (ECCN)     | EAR99                                                 |
| REACH Compliance                                | REACH (EC 1907/2006) (06/27/2024, 241 SVHC) Compliant |
| RoHS Compliance                                 | RoHS 3 (2015/863/EU) Compliant                        |
| Schedule B Number                               | 841459.9080                                           |
| U.S.–Mexico–Canada Agreement (USMCA) Qualifying | No                                                    |

Cool down electronics in areas that require frequent and intense sanitizing. Rated IP69K, these AC-powered fans withstand harsh jets of water and block dust from reaching their motor. All have ball bearings that reduce friction and heat from operation, extending their life. Also known as muffin fans.



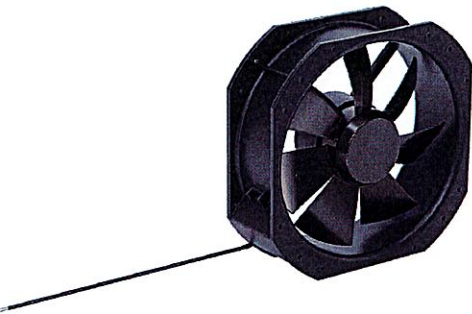


The information in this 3-D model is provided for reference only.



High-Pressure Washdown Equipment-Cooling Fan

115V AC, 280 mm High x 280 mm Wide x 89 mm Deep Overall

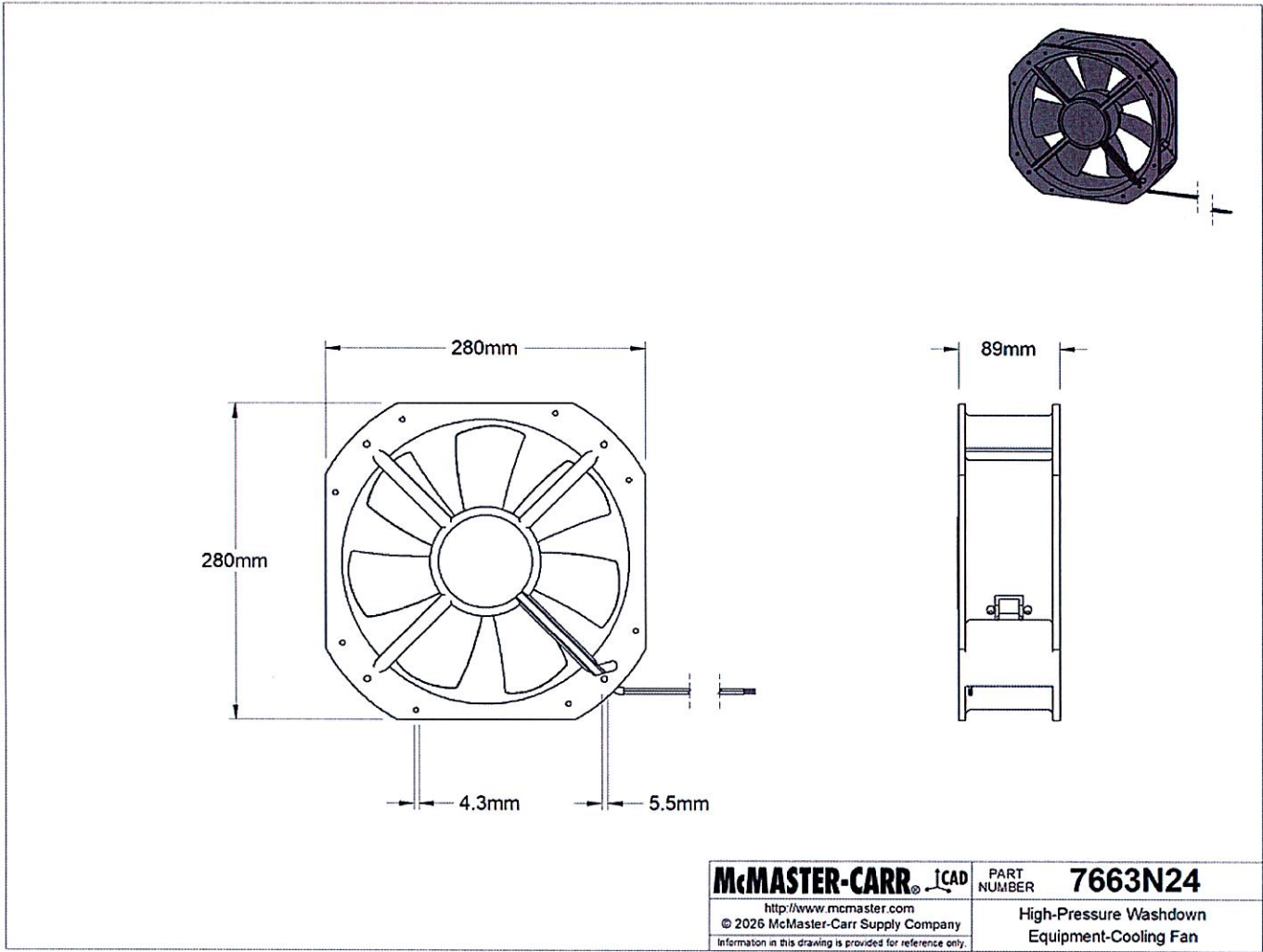


Delivers tomorrow

\$305.41 each  
7663N24

|                                                 |                                                       |
|-------------------------------------------------|-------------------------------------------------------|
| Shape                                           | Oblong                                                |
| Frame Material                                  | Aluminum                                              |
| Blade Material                                  | Steel                                                 |
| Voltage                                         | 115V AC                                               |
| Electrical Phase                                | Single                                                |
| Frequency                                       | 60 Hz                                                 |
| Electrical Connection                           | Wire Leads                                            |
| Overall                                         |                                                       |
| Height                                          | 280 mm                                                |
| Width                                           | 280 mm                                                |
| Depth                                           | 89 mm                                                 |
| Airflow                                         | 1,300 ft <sup>3</sup> /min                            |
| Volume (Sound)                                  | 73 dBA                                                |
| Current                                         | 1.75 amp                                              |
| Mounting                                        |                                                       |
| Fasteners Included                              | No                                                    |
| Hole Diameter                                   | 4.3 mm, 5.5 mm                                        |
| Enclosure Rating                                | IP69K                                                 |
| Certification                                   | CE Marked, UL Recognized Component                    |
| Application                                     | Cooling                                               |
| Environment                                     | Dusty, High-Pressure Washdown, Washdown, Wet          |
| Power Source                                    | Electric                                              |
| Thermostat Included                             | No                                                    |
| Country of Origin                               | Taiwan                                                |
| DFARS Compliance                                | Specialty Metals COTS-Exempt                          |
| Export Control Classification Number (ECCN)     | EAR99                                                 |
| REACH Compliance                                | REACH (EC 1907/2006) (06/27/2024, 241 SVHC) Compliant |
| RoHS Compliance                                 | RoHS 3 (2015/863/EU) Compliant                        |
| Schedule B Number                               | 841459.9080                                           |
| U.S.–Mexico–Canada Agreement (USMCA) Qualifying | No                                                    |

Cool down electronics in areas that require frequent and intense sanitizing. Rated IP69K, these AC-powered fans withstand harsh jets of water and block dust from reaching their motor. All have ball bearings that reduce friction and heat from operation, extending their life. Also known as muffin fans.



The information in this 3-D model is provided for reference only.





## Wall-Mount Jet-Stream Fan

120V AC, 10" Blade Diameter

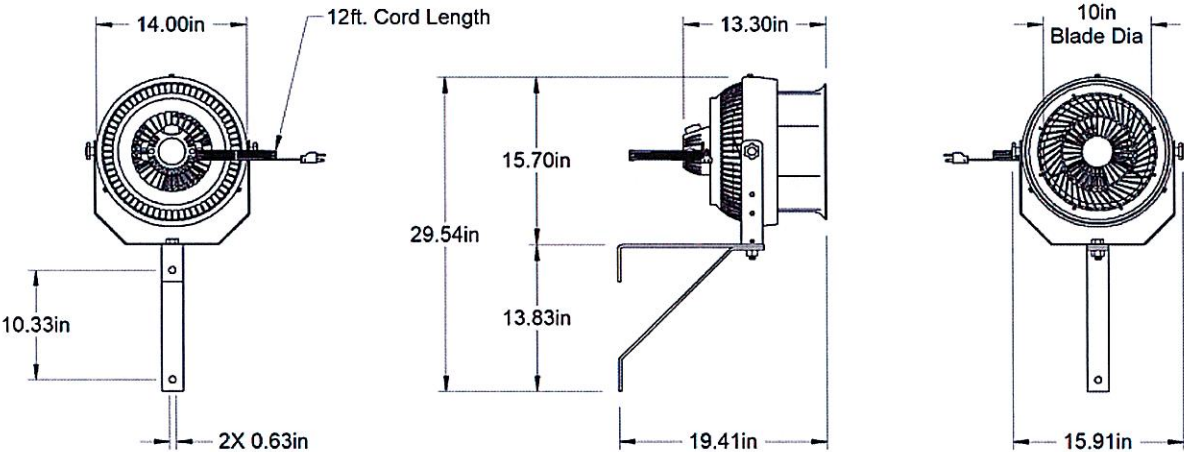


Delivers tomorrow

\$438.60 each  
2138K12

|                                                 |                                        |
|-------------------------------------------------|----------------------------------------|
| Motor Enclosure Type                            | Totally Enclosed                       |
| Voltage and Connection                          | 120V AC, Single Phase—Three Blade Plug |
| Fan Head Movement                               | Non-Oscillating                        |
| Blade                                           |                                        |
| Diameter                                        | 10"                                    |
| Material                                        | Nylon                                  |
| Airflow                                         | 1,500 ft <sup>3</sup> /min             |
| Volume                                          | 56 dBA @ 15 ft.                        |
| Motor Speed                                     | 1,600 rpm                              |
| Number of Speeds                                | 2                                      |
| Power                                           | 1/8 hp                                 |
| Current                                         | 0.8 amp                                |
| Overall                                         |                                        |
| Height                                          | 16"                                    |
| Width                                           | 16"                                    |
| Depth                                           | 13"                                    |
| Cord Length                                     | 12 ft.                                 |
| Drive Type                                      | Direct                                 |
| Housing Material                                | Polypropylene                          |
| Guard Material                                  | Polypropylene                          |
| Mounting                                        |                                        |
| Location                                        | Wall                                   |
| Fasteners Included                              | Yes                                    |
| Hole Diameter                                   | 5/8"                                   |
| Type                                            | Screw On                               |
| Application                                     | Cooling                                |
| Mobility                                        | Stationary                             |
| Power Source                                    | Electric                               |
| Country of Origin                               | United States                          |
| DFARS Compliance                                | Specialty Metals COTS-Exempt           |
| Export Control Classification Number (ECCN)     | EAR99                                  |
| REACH Compliance                                | Not Compliant                          |
| RoHS Compliance                                 | Not Compliant                          |
| Schedule B Number                               | 841451.0090                            |
| U.S.–Mexico–Canada Agreement (USMCA) Qualifying | No                                     |

Often used to cool warehouse aisles and production lines, these fans generate a stream of air that travels farther than air from conventional fans. They have a direct-drive motor, which has fewer moving parts than belt-drive motors for less maintenance. They also have a totally enclosed motor enclosure, which can be used in dirty, dusty, and wet environments.



Electrical Connection: Plug

|                                                                                                             |                |                               |  |  |
|-------------------------------------------------------------------------------------------------------------|----------------|-------------------------------|--|--|
| <b>McMASTER-CARR®</b>  | PART<br>NUMBER | <b>2138K12</b>                |  |  |
|                                                                                                             |                |                               |  |  |
|                                                                                                             |                |                               |  |  |
| <a href="http://www.mcmaster.com">http://www.mcmaster.com</a>                                               |                | Wall-Mount Jet-<br>Stream Fan |  |  |
| © 2026 McMaster-Carr Supply Company                                                                         |                |                               |  |  |
| Information in this drawing is provided for reference only.                                                 |                |                               |  |  |

The information in this 3-D model is provided for reference only.